# Woosik Kim

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## EXPERIENCE

**Match-a** **Jun 2026 - Present**
*Software Engineer*
- Lead **3** software engineers building a mobile and web social app, including a feature that renders a user's face as a privacy-preserving avatar.
- Engineered the GitHub Actions CI pipeline gating every merge on type checks, linting, a **340+** test suite, and a build.
- Hardened security with Twilio webhook signature checks, LLM route rate limits, and Semgrep and gitleaks scans in CI.
- Instrumented Sentry error monitoring across server, edge, and client runtimes covering **80+** API routes.

**LG Uplus** **Jun 2026 - Jul 2026**
*AI Red Team Intern, Information Security Center*
- Architected a **7-stage** AI red team lifecycle for a major South Korean telecom carrier to secure Shadow AI company-wide.
- Defined AI asset discovery across **5** detectors (DNS, EDR, OAuth consent, secret scans, cloud audit logs).
- Built a **19-scenario** telecom attack taxonomy mapped to MITRE ATLAS and OWASP LLM Top **10**, verified with canary tokens.
- Designed AVS, a CVSS-based AI risk score adding agency and exploit rate, and defended it to the CTO and security chief.

**Youngduk Electronics** **May 2017 - May 2020**
*Operations Technology Assistant*
- Built the internal logistics and inventory management program for an electronic components distributor, designing its data model and reporting flow.
- Automated inventory tracking and ERP reporting through it, cutting stock errors **30%** and raising data accuracy **20%**.
- Maintained and updated the program over **3** years as the product catalog and reporting requirements changed.

## RESEARCH & PROJECTS

**ShowBrand: AI Reliability Research** | Faculty Mentor Program, UC San Diego **Jan 2026 - May 2026**
*Advised by Prof. Pengtao Xie (ECE)* | *Presented at MURALS 2026* | github.com/woosik-study/ShowBrand-Product
- Originated the project and ran **4** controlled experiments (baseline, blur, logo occlusion, AI manipulation) on **136** images.
- Found an overconfidence failure: with logos occluded, brand accuracy fell **17 points** while model confidence fell only **0.7**.
- Built a Next.js app on Claude vision and a Python harness (pandas, Pillow) to degrade, run, and score test images.
- Shipped the findings into the app as confidence thresholds, a low-confidence retake prompt, and risk-based wording.

**Coverage Keeper** | AI Medicaid Letter Reader | 2nd Place, Healthcare Track, Solix Empower Hackathon **Oct 2026**
*Team of 3* | *Next.js, TypeScript, Anthropic API, zod* | github.com/AdityaJadhav17/coverage-keeper
- Engineered the letter-reading backend: vision LLM calls with schema-validated output, **429/5xx** retry, and a safe fallback.
- Implemented an upload API with type, size, and rate limits, output guardrails, and no letter content in logs.
- Curated a **13-notice** labeled eval set with a fixed dev/test split, reading **4 of 4** held-out deadlines correctly.

**WatchTower** | Self-hosted Browser Monitoring SDK (Team Project) **Apr 2026 - Jun 2026**
*Live demo:* cse110-sp26-group09.github.io/Watchtower-Course-Project
- Hardened the Node.js event API with request validation, **37+** server unit tests, and Playwright end-to-end tests.
- Fixed a stored XSS vulnerability and added Beacon API delivery and Navigation Timing metrics to the browser SDK.

## EDUCATION

**University of California, San Diego** **Expected Dec 2027**
*Bachelor of Science in Computer Science*
**Relevant Coursework:** Software Engineering, Computer Systems, Statistical Learning, Recommender Systems, AI & Search

## TECHNICAL SKILLS

**Languages:** Python, TypeScript/JavaScript, C/C++, SQL
**Web & Backend:** React, Next.js, Node.js, REST API design, PostgreSQL (Supabase)
**AI & ML:** LLM APIs (Anthropic, Vertex AI/Gemini), vision language models, structured output, LLM evaluation, uncertainty analysis
**Security:** AI red teaming, threat modeling, MITRE ATLAS, OWASP Top **10** / LLM Top **10**, SAST (Semgrep), secret scanning (gitleaks)
**DevOps & Tooling:** Linux, bash shell, Git, GitHub Actions CI/CD, Vercel, Sentry, Playwright

## LEADERSHIP, ACTIVITIES & AWARDS

**Military Service:** Team Leader, Republic of Korea Army (**2020** - **2022**), led and trained a unit of **36** recruits
**Activities:** Web Developer, TESC | Developer, Likelion UCSD | IDEA Scholar, Jacobs School of Engineering | Member, Council on Undergraduate Research (CUR) | Math & Programming Mentor, MESA
**Awards:** STEM Math Achievement Award (**2025**), Bill and Mary Kim Scholarship (**2024**, **2025**) | **Languages:** English, Korean